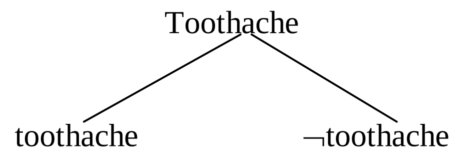
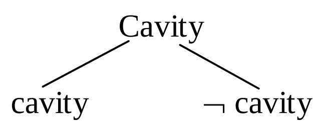
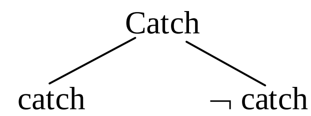
	toothache		¬toothache	
	catch	¬catch	catch	¬catch
cavity	0.108	0.012	0.072	0.008
¬cavity	0.016	0.064	0.144	0.576







$P(\text{cavity} \wedge \text{catch})$ = P('The patient visiting the doctor has cavity and catch.') =

$P(\text{toothache} \wedge \textbf{cavity} \wedge \textbf{catch}) + P(\neg\text{toothache} \wedge \textbf{cavity} \wedge \textbf{catch}) = 0.108+0.072 = 0.18$

Features:

	toothache		¬toothache	
	catch	¬catch	catch	¬catch
cavity	0.108	0.012	0.072	0.008
¬cavity	0.016	0.064	0.144	0.576

1. $P(\text{toothache} \wedge \text{cavity} \wedge \text{catch}) + P(\text{toothache} \wedge \text{cavity} \wedge \neg\text{catch}) + \ldots$
   $+ P(\neg\text{toothache} \wedge \neg\text{cavity} \wedge \neg\text{catch}) = 0.108 + 0.012 + \ldots + 0.576 = 1.$

	toothache		¬toothache	
	catch	¬catch	catch	¬catch
cavity	0.108	0.012	0.072	0.008
¬cavity	0.016	0.064	0.144	0.576

2. $P(\text{toothache} \wedge \text{cavity}) = 0.108 + 0.012 = 0.12$
   $P(\text{toothache} \vee \text{cavity}) = 0.108 + 0.012 + 0.016 + 0.064 + 0.072 + 0.008 = 0.28$
   $P(\neg\text{catch}) = 0.012 + 0.064 + 0.008 + 0.576 = 0.66$
   …

	toothache		$\neg$toothache	
	catch	$\neg$catch	catch	$\neg$catch
cavity	0.108	0.012	0.072	0.008
$\neg$cavity	0.016	0.064	0.144	0.576

3. $P(a|b) = P(a \wedge b) / P(b)$, where $P(b) > 0$.
   [P(a|b) – Probability of the truth of the proposition a, given the truth of the proposition b]

   $P(cavity \mid toothache) = P(cavity \wedge toothache) / P(toothache) =$
   $(0.108+0.012) / (0.108+0.012+0.016+0.064) = 0.12/0.2 = 0.6$

	toothache		$\neg$toothache	
	catch	$\neg$catch	catch	$\neg$catch
cavity	0.108	0.012	0.072	0.008
$\neg$cavity	0.016	0.064	0.144	0.576

4. $P(toothache \wedge \neg cavity) = 0.016 + 0.064 = 0.08$
   $P(toothache \mid \neg catch) = P(toothache \wedge \neg catch) / P(\neg catch) = (0.012+0.064)/0.66 =$
   $0.076/0.66 \approx 0.115$

   Homework: $P((toothache \wedge \neg cavity) \mid catch) = ?$
   $P(catch \mid (toothache \wedge \neg cavity)) = ?$
   $P(catch \mid (\neg toothache \vee cavity)) = ?$
   … .